Design FMEA - VBF-T6LC-DFMEA-01

Design FMEA (qualitative, simulation-signal based) - VBF-T6LC-DFMEA-01
Qualitative version based on simulation signals; complete process/supplier FMEA is N/A (beyond pure-simulation boundary).

	Failure cause	Simulation signal (detectability)	Severity	Occurrence	Design-side mitigation
ge)	anode potential < 0 V at 4C	`anode_potential_v` min +0.0894 V (margin 0.089 V)	High	Low	N/P raised to 1.68 (110 umm an
)	4C heat generation exceeds cooling	`T_max_K` 322.59 K vs 323.15 K (margin 0.56 K)	High	Low	cooling h = 45 W/m2/K
window)	4.7 V upper cut-off	not directly simulated (true DFT endorsement skipped, real_compute=false)	Medium	Medium	high-voltage electrolyte/additive
	thin electrodes / high overhead	ED 1005.8 Wh/L vs 950 (margin 5.9%)	Medium	Low	thin separator/CC reclaim volum
	high 4.7 V charge drives anode SEI	SEI 233 nm (SPMe) vs 500 nm	Medium	Low	anode coating k_SEI = 1e-15 m/

Conclusion
Highest-risk item: electrolyte oxidative decomposition at 4.7 V (unverified in this virtual run - true DFT endorsement skipped).
Mitigations implemented for plating, temperature, and SEI; the complete process/supplier FMEA is annotated N/A.